



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2011/2012
INSTITUTION BASED PROGRAMME (DECEMBER SESSION)
EXAMINATION FOR POSTGRADUATE DIPLOMA IN EDUCATION
ECT/PG/9: CHEMISTRY METHODS

DATE: FRIDAY, 4TH MAY 2012

TIME: 4.30 P.M. – 6.30 P.M.

Instructions: Answer question 1 and any other TWO

Q1ai) We teach alkenes as a homologous series at high school. Write out the first three alkenes in the homologous series.

ii) What concept in regard to molecular formulae of alkenes would you tell your students comes out of the first three compounds in this regard?

bi) The high point of teaching is problem solving. What is the role of the teacher and the student respectively in problem solving?

ii) In regard to (aii) above a student gives you the wrong answer. What would you say to the student?

ci) We teach for transfer of learning. With an example of chemistry illustrate lateral transfer of learning.

ii) Similarly illustrate vertical transfer of learning.

di) "Repetition and Pavlovian psychology have their place in the learning/teaching of chemistry"

You are teaching the first four members of the alkane series. How would you see the relevance of this assertion in your class?

ii) Is there an exception? How would you explain it?

Q2 ai) It is important that in teaching chemistry we form a link to what happens world-wide. A student asks you, "Iraq was invaded for allegedly having chemical weapons. Can you give me two examples of chemical weapons?" What would you say. (Give industrial or chemical names)

ii) The same student follows up, "I am told that chemical weapons have to meet certain physical and chemical characteristics besides killing in large numbers. Kindly tell me two." What would you say?

bi) As a teacher you double up as a laboratory manager. What would you say to your technician who has a habit of using tap water to prepare bench solutions?

ii) Give three reasons why you would use alphabetical labeling in your laboratory.

Q3ai) Accidents will happen in the learning/teaching of chemistry. How would you treat a Student who spills sodium hydroxide on their skin?

ii) During the same lesson another student breaks glass into their skin. What do you do ?

bi) International issues are part of teaching chemistry. What would you tell your students the Kyoto Protocol is about?

ii) Similarly why did the USA refuse to ratify it?

Q4ai) Measurement and testing are part of our teaching. Consider this problem that links biology and chemistry.

“ Vitamin C has the formula $C_6H_8O_6$. Calculate the number of molecules of vitamin C in 250 mg of vitamin C”.

C= 12.00; H=1.00; O=16.00; Avogadro's Constant = 6.02×10^{23}

Give the units of C, O, H, and the Avogadro's Constant.

ii) The magnitude of the Avogadro's constant is important in chemistry. What would you tell your students that this represents.

b) Solve for the problem clearly and throughout showing the part of the units as you would to your class.

Q5ai) You may be consulted in the construction of a laboratory in your school. Why would you recommend that it faces E-W orientation?

ii) We anneal glass tubing to desired shapes for our experiments. What procedure should we follow?

bi) A student asks you, ' Sir/Madam could you please write for me the name and formula of a diprotic and a triprotic acid?' What would you write?

ii) The same student follows up "If we had to write out a formula for an unknown polyprotic acid what would it be?" What would you write.
